

NadiRakshak AI (■■■■ ■■■■■■)

Pre-Monsoon Drainage Choke & Flood Inundation Proof-of-Work Auditor

A Proactive Civic Climate Resilience Platform Built with Gemini 1.5 Flash & Google Cloud

Track: Clean Air & Climate Resilience	Team Leader: Anmol Suwalka
Tech: Gemini 1.5 Flash, Vertex AI, Earth Engine	Team Member: Tejasvi Gautam

1. The Problem: The ■500+ Crore Pre-Monsoon Drainage Blindspot

Every monsoon, Indian cities drown in severe urban flash floods, paralyzing livelihoods and causing deadly post-flood epidemics (Dengue, Cholera). **The root cause is not just rainfall—it is unverified stormwater drain desilting.**

1. Zero Proof-of-Work Verification

Municipalities award ■300-800 Cr in desilting tenders. Contractors clean only the first 5 meters of a culvert, dump sludge by the roadside, and submit fraudulent paper bills.

2. Subterranean Silt & Rubble Choking

Construction slurry, debris, and plastic waste accumulate beneath road culverts completely invisible to cursory surface inspections until 50mm of rain causes massive backflow.

3. Purely Reactive Disaster Response

Emergency machinery is only mobilized after citizens are knee-deep in water and helpline lines are overwhelmed. No system predicts which culverts will choke 48h before the downpour.

2. The Solution: NadiRakshak AI Overview

NadiRakshak AI shifts municipal disaster governance from **reactive emergency rescue to proactive infrastructure auditing** using smartphone cameras and Google AI:

Gemini 1.5 Multimodal Silt Vision Evaluates cross-sectional drain capacity, silt depth, and solid waste choking (0-100%) from ordinary phone photos in under 400ms. Zero sensor hardware cost.	Anti-Fraud Verification Engine Uses multimodal embeddings to compare contractor before/after submissions. Detects duplicate reused photos and superficial surface skimming.
Predictive Runoff & Flood Backflow Couples Google Earth Engine elevation models with IMD rainfall radar to forecast exactly which wards will suffer backflow 48h in advance.	MP / Collector Action War Room Ranks priority culverts into an actionable dispatch queue for municipal backhoes (JCBs) and suction trucks before clouds gather.

3. Technical Architecture & Google Tech Stack

Designed for enterprise reliability, high-throughput citizen auditing, and zero municipal maintenance overhead:

Component	Google Technology	Core Role & Function
AI Vision Core	Gemini 1.5 Flash (Vertex AI)	Cross-sectional hydraulic analysis, silt depth estimation & duplicate photo fraud detection.
Spatial Inundation	Google Earth Engine	High-resolution Digital Elevation Models (DEM) for hydrological surface runoff simulation.
Serverless Backend	Google Cloud Run	Autoscaling microservices handling photo ingestion, contractor billing audits & telemetry.
Maps & Geocoding	Google Maps Platform	Interactive culvert choke clustering, municipal GPS dispatch routing & ward visualization.
Alert Distribution	Firebase Cloud Messaging	Sub-second push notifications & SMS gateway triggers for vulnerable slum clusters.

4. Deep Dive: Zero-Hardware Silt & Fraud Audit

How Gemini 1.5 solves what multi-lakh ultrasonic flow sensors fail to do in muddy Indian storm drains:

Input: Single Photo

Ward sanitation staff or citizens snap a photo with an entry-level smartphone. No physical probe, no calibration.

Multimodal Spatial Analysis

Gemini 1.5 Flash measures the visible culvert rim, water meniscus, sediment buildup, and debris composition (concrete rubble vs. plastic bottles).

Proof-of-Work Verification (Anti-Fraud)

Verifies if contractor genuinely cleared the bed to required design depth. Compares image hash & shadow angles against previous years to eliminate fake duplicate bills.

Automated Governance Output

Generates an instant audit report: Silt % | Flow Capacity Status | Contractor Bill Recommendation (Approve / Penalty Notice).

5. 48-Hour Pre-Rain Inundation Simulation

Combining physical hydrology with meteorological radar to beat the storm:

IMD Radar Precipitation Coupling Tracks incoming monsoon cloudbursts in 6-hour forecast intervals (e.g. 72mm in 36h).	Hydraulic Flow Deficit Calculation Calculates drainage discharge capacity (Cubic Meters/sec) based on audited silt choke %.
Backflow Colony Warning Maps which residential colonies and low-lying slums will submerge when drain trunk reaches 100% capacity.	Pre-emptive Asset Placement Moves high-capacity diesel dewatering pumps to at-risk underpasses before roads flood.

6. The MP & District Collector War Room Console

Giving elected leaders and district commissioners real-time command over infrastructure:

Executive Priority Queue: Ranks 500+ ward culverts by population risk and silt choke ratio. No guesswork.
Autonomous Contractor Penalty Notice: Auto-drafts legal penalty letters to defaulting contractors when desilting is missing.
Geo-Tagged Machinery Dispatch: Dispatches municipal JCBs and suction tankers with verified GPS arrival tracking.
Constituency Report Card: Gives the MP clear proof of pre-monsoon governance preparedness to share with citizens.

7. Social Impact & Alignment with National Missions

Directly delivers on India's urban resilience and sustainability benchmarks:

Metric / Pillar	Projected Impact	National Scheme Alignment
Flood Damage Prevented	■30-50 Crore per urban constituency per monsoon	NDMA Urban Flood Guidelines
Tender Leakage Eliminated	100% verifiable proof-of-work before bill release	AMRUT 2.0 & Municipal Transparency
Public Health Protection	Substantial reduction in post-flood vector-borne outbreaks	National Health Mission (NHM)
Marginalized Wards Shielded	Protects low-lying informal settlements and slums	SDG 11 (Sustainable Cities & Communities)

8. Unique Selling Proposition & Scalability

Why NadiRakshak AI stands out in the competition:

- 1. Proactive vs. Reactive:** Stops disasters before clouds gather, rather than deploying inflatable rescue boats after homes submerge.
- 2. Zero Hardware Barrier:** Turns any basic Android smartphone into an industrial-grade hydraulic audit device.
- 3. Anti-Fraud Native:** Directly addresses the multi-crore corruption and negligence in municipal drainage contracts.
- 4. Nationwide Scalability:** Modular containerized architecture ready for deployment across 500+ Indian Municipal Corporations.

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Transforming Municipal Drainage from a Corrupt Blindspot into an AI-Shielded Climate Fortress

Built for: Code for Communities 2.0 (Google Cloud & GDG India)

Track: Clean Air & Climate Resilience

Live Prototype: Interactive Web Dashboard with Gemini 1.5 Multimodal Silt Vision

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